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Comparative study analysis of practical EEG sensors in medical diagnoses

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ABSTRACT

EEG records the brain's electrical function. It displays the difference in voltages from distinct sites of the brain over a period of time. It plays a major role in understanding the brain. It also helps to diagnose disorders related to sleep, and find the epileptogenic foci in the brain. Recording accurate EEG signals, however, comes with its own limitations. Gels applied to the scalp are abrasive in nature. Gels dry up over time, and also irritate the skin. There are different kinds of electrodes that record EEG signals with appreciable accuracy, keeping comfort and practicality in mind. Some of them are: dry electrodes, capacitive electrodes, electrodes with nanomaterials. The electrodes mentioned above are discussed in this paper. It discusses the advancement in the field of EEG. This paper shows the ease at which signals can be captured, using the aforementioned electrodes. Cheaper, more accessible EEG electrodes are making it possible to bring affordable healthcare outside the confines of the hospitals and clinics.

1. Introduction

Recording brain activity opens doorways to different diseases and conditions, and how to cure [1] them. There are different techniques for monitoring the brain activity, such as [1]:

- fMRI (functional Magnetic Resonance Imaging)
- PET(Positron Emission Tomography)
- CT(Computed tomography)
- MEG(Magnetoencephalography)
- EEG(Electroencephalography)

When feasibility, accessibility and affordability are taken into consideration, it is clear that EEG is the promising candidate for recording signals from the brain. EEG is comparatively safer for the subjects as they are not introduced to external magnetic fields or electrical fields.

There are various methods to measure the electrical signal produced by the brain:

- Electrocorticogram (ECoG)
- Single Unit Recording and Multi-unit recording
- Local Field Potential (LFP)
- Electroencephalogram (EEG)

Richard Caton, a British physiologist, first reported detecting electrical waves in the cerebral cortex. He submitted this in the British Medical Journal (1875). Hans Berger (German psychiatrist) published the encephalogram. Brain's activity can be recorded from the surface of the cerebral cortex. This phenomenon was first reported by Woldring and Dirken in 1950 [2].

The first recording devices were rudimentary. The patients had silver wires inserted under their scalps. Next, they attached the silver foil to the patient's head with the help of rubber bandages. Highly sensitive galvanometers like the Siemens double-coil recording galvanometer [3] were used to measure very small voltage values (1/10000th of a volt). This played a key role in capturing the signals produced by the brain.

Berger saw that EEG wave diagrams could act as indicators of brain diseases. The waves produced by a diseased brain were different from the waves produced by a healthy brain. EEG thus expanded the realm of possibilities in understanding the brain more intuitively.

The EEG or the electroencephalogram is a widely used non- invasive method for checking and monitoring the brain. The entire method is based on placing conductive electrodes on the scalp which measure the potential that arises outside the confines of the skull (due to the firing of neurons in the brain). In the traditional method, about two or more dozen electrodes are placed on standard positions of the scalp [4]. The voltage fluctuations (in the order of microvolts) are then amplified a couple of thousand times. The postsynaptic potential is the reason behind the extracellular electric field generated [4]. EEG can provide deep insight into the inner workings of the brain as it gives outputs that have a high temporal resolution. Neurobiological research and medical diagnoses rely on EEG. For example, studies have indicated that space func-

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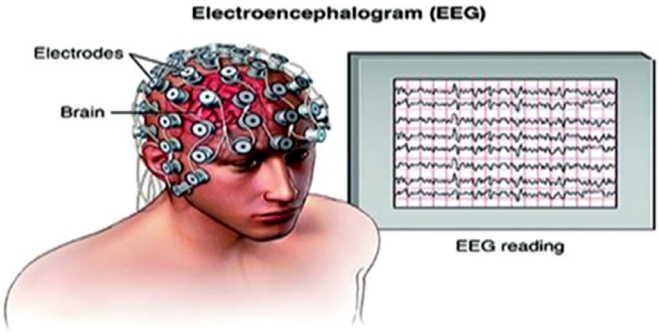


Fig. 1. Illustration of EEG recording [6].

tional connectivity mapping of the brain(with EEG) helps in managing brain disorders like epilepsy [5] (Figs. 1–20).

EEG electrodes are made of conductive materials that measure the electrical potential difference between the bioelectric field from the scalp and its relative reference point. The voltage can be detected on the scalp only when thousands of neurons fire together. The collective activation of the neurons in a neural mass (which consist of 10^4 to 10^7 neurons) is called synchronism [4]. The result of these groups of cells getting excited several times synchronously is a large number of rhythmic waves as the output. The sensors measure signals that emanate from the brain, these signals vary with time.

Hence, these signals are called time-varying signals. It records the electrical potentials of the brain by gauging the electric fields that emanate from the scalp. EEG has low spatial resolution and high temporal resolution. It records these changes (in the cortex) in the order of milliseconds. The signals are recorded within fractions of a second of them occurring. This is also one of the reasons why it is preferred over other methods (fMRI and PET), even though EEG has low spatial resolution. The measurement relies on direct, low-resistance contact between the sensor and the scalp [9]. Gels usually act as a medium between the sensor and the patient's scalp.

This is where problems come up; applying the gel takes quite a lot of time, it can spread through the subject's hair and inadvertently cause the electrodes to short. It can also dry over time. They act as an irritant and are often uncomfortable. The gels are abrasive. They can cause allergic reactions and change characteristics with time. This paper discusses the alternative methods of achieving reliable EEG outputs

1.1. Related works

W. Tatum [10] goes in depth about what electrodes are and how they aid in recording signals. Recording electrodes are used to provide an interface between biological tissue and the metal components that make up the electrode. It throws light on the physiology of cerebral potentials. Pyramid cells are considered to be the important contributors to the biopotentials that make up EEG.

L.-D. Liao *et al.* [11] talks about a dry contact sensor without the need for elaborate skin preparation. This paper talks about the advantages of dry EEG sensors, specifically about the long term impedance measurements that makes it a viable alternative for electrodes.

J. W. Y. Kam *et al* [12] is about a dry electrode that can help in presurgical monitoring of seizures in the comfort of the patients' home. It demonstrated how the signal quality is comparable with that of the conventional wet electrodes. Comprehensive analysis of different aspects of EEG is shown.

Y.-H. Chen *et al.* [13] is a paper on flexible polymer-based electrodes that can be used to measure EEG signals comfortably. Electrodes with around 45% carbon content were used to detect signals from the brain as it is the best trade-off between electrical and mechanical properties. The paper concluded that the polymer electrode is a promising alternative to conventional gel electrodes.

S. Lee, *et.al.* [14] is a paper about an electrolyte-free dry electrode that merges spring-loaded probes with an active buffer circuit to increase the input impedance. The proposed method in the paper boosts impedance roughly to 147.5% of the original value within the given EEG bandwidth.

M. N. Barshutina *et al* [15] talks about biocompatible CNT hybrid composites that are flexible and highly conductive. The paper talks about a non-toxic and reliable technique that can be the future of wearable electronics. This method is affordable and scalable.

B. Unnadkat *et al* [16] talks about the advantages of capacitive electrodes: there are no leakage-currents or electrical shorts. Using capacitive electrodes is safer when compared with the other methods as there is no contact between the subject and the electrode. Inconspicuous long term monitoring is possible with capacitive electrodes. The advantage of this method is that it requires no complex medical procedures for set up.

All the papers cited here go in depth about the viable substitutes for the conventional wet electrodes. In each of these papers, several important comparisons are drawn between the wet electrodes and the prototypes introduced.

In each of the cases it is shown that the prototype electrodes perform sufficiently well enough to replace the wet electrodes This paper gathers information about similar methods to come to a conclusion about a particular kind of electrode, and how it can replace the wet electrodes.

2. Methods of acquiring EEG outputs

2.1. Dry contact sensors

Dry electrodes were introduced to overcome the shortcomings of wet electrodes

Ruffini *et. al.* [17] invented dry electrodes which consist of an array of multi-walled CNTs with Ag/AgCl coating [2]. This type of electrode penetrates the skin. However, non-invasive electrodes are preferred to invasive electrodes. Here are a few examples: a copper comb-like electrode in direct contact with the scalp [18].

Dry contact EEG sensors are one of the simplest methods to obtain EEG signals. One of the disadvantages of contact sensors is that they have high contact impedance. The electrodes in this paper can detect alpha rhythms with low noise of 1.131 μVRMS [14].

Flexible electrodes can be made from polymers [13]. Their flexibility is an advantage as it moulds with the curves of the scalp, making it easier for them to measure the signals.

Micro-electro-mechanical systems or MEMS replaced the conventional electrodes. Dry MEMS EEG electrodes [11] have many pointed edges on the surface that come in contact with the skull to acquire signals. The contact probes make it easier for the substrate to mould in with the scalp's surface [11]. However, there is no persuasive evidence about the signal quality under different sites (hairy and hairless) using the electrodes mentioned above [11].

This method has the following properties:

1 The first advantage is that they can operate without the use of conductive gels
2 The impedance offered by the dry electrodes is lesser than the impedance offered by wet electrodes when used on hairy sites.
3 It is possible to mass-produce dry electrodes (it is interesting to note that the cost per unit will decrease with the increase in the quantity of the sensors manufactured.)
4 The spring contact probes are capable of buffering the electrode against the applied force [11].

Dry electrodes provide stable and reliable EEG signals. The signal quality of the EEG sensors do not deteriorate over time. According to

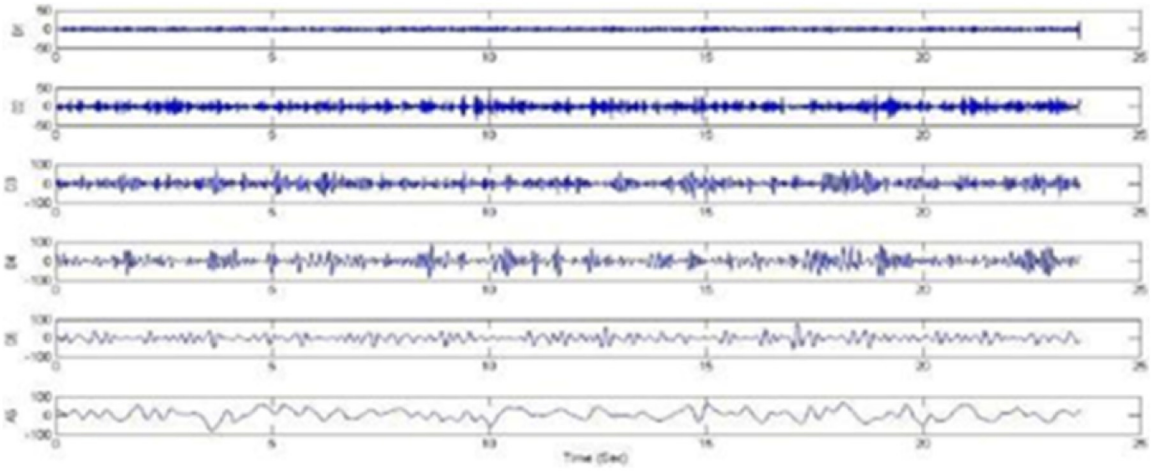


Fig. 2. Wavelet output for normal EEG signal [7].

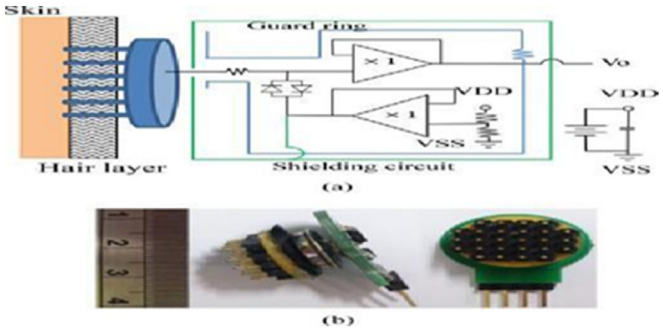


Fig. 3. Comb-like electrodes(dry) to measure EEG in hairy sites [18].

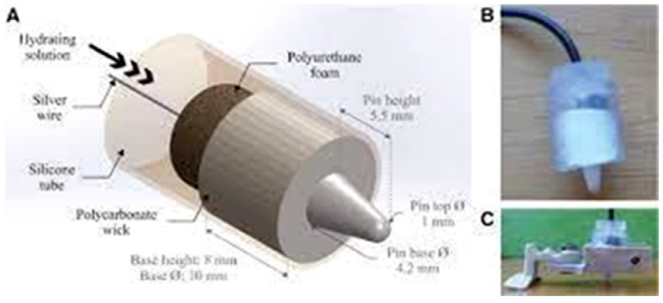


Fig. 4. Quasi dry electrode [19].

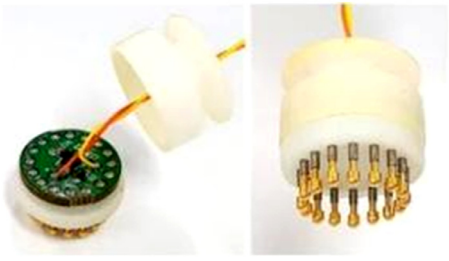


Fig. 5. The proposed active dry electrode consists of a given unit and the auxiliary circuit board for power supply [14].

a study performed by *Julia W.Y Kam* [12] dry electrodes and wet electrodes showed good correlation($r = 0.54–0.89$) when amplitude, topography and low frequency spectral power were taken into consideration. All these parameters contribute to the possibility of these electrodes being the future of EEG sensors.

2.2. Textile based sensor

Methods like the Cerebral Function Monitoring(CFM) [21] employ compressed single channel [12] amplitude-integrated electroencephalography. The data is recorded with the help of two biparietal electrodes [21]. These methods rely on a comparatively less number of channels and display compact versions of EEG [28]. Brain activity like seizures can be localised to one point. As CFM uses only a few electrodes,

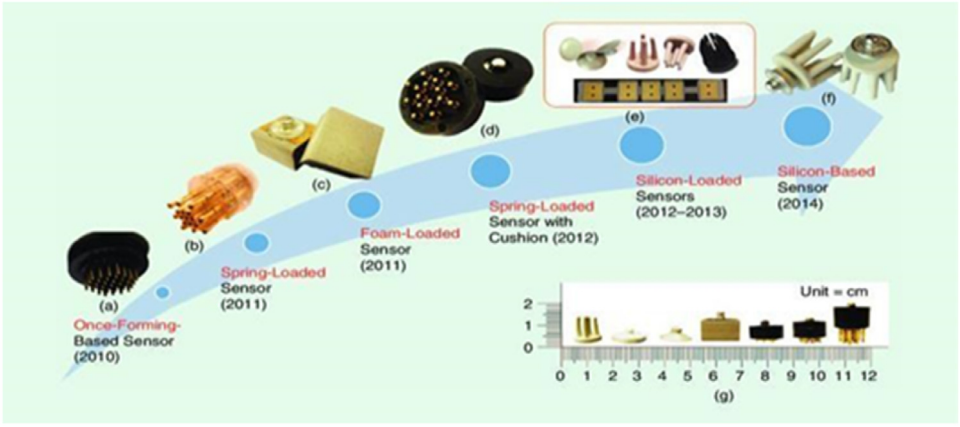


Fig. 6. Dry contact sensors that measure EEG signals on the scalp. Evolution over the years [20].

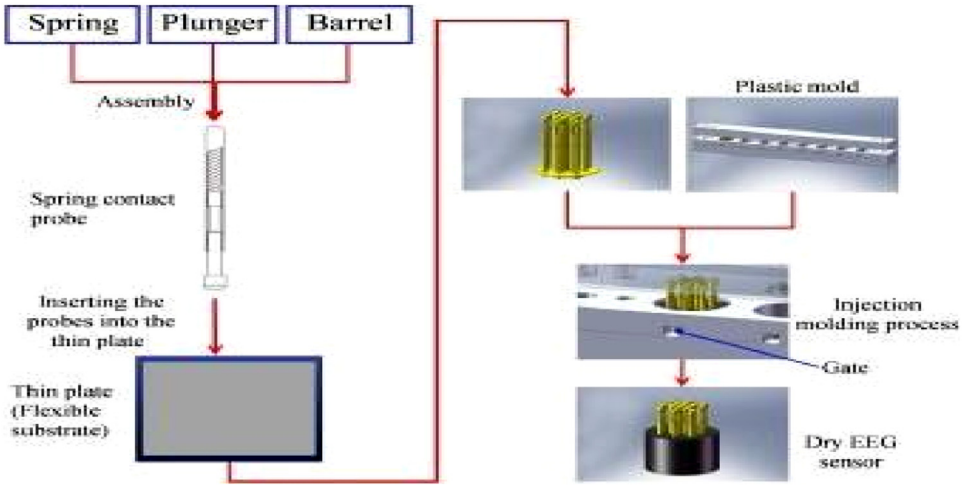


Fig. 7. The fabrication process of novel dry EEG sensors [11].

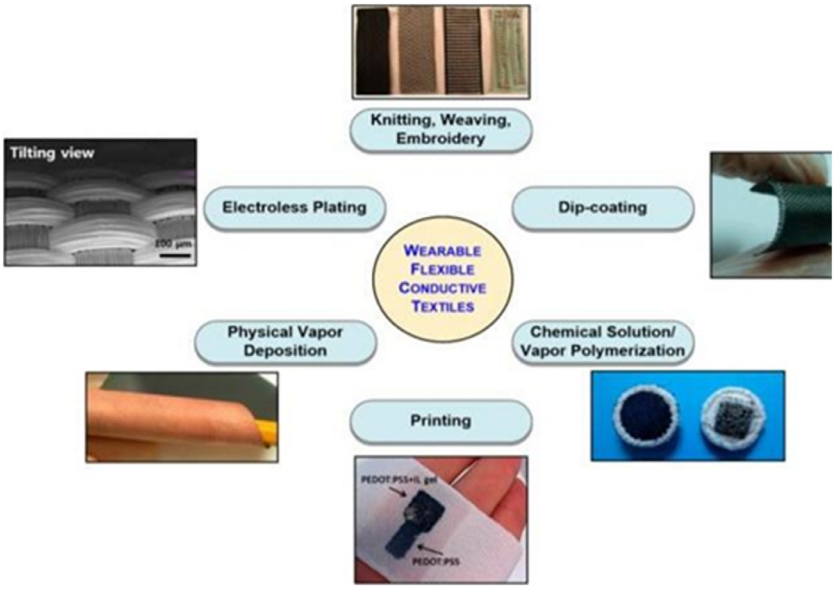


Fig. 8. Primary methods of realizing conductive textiles [25].

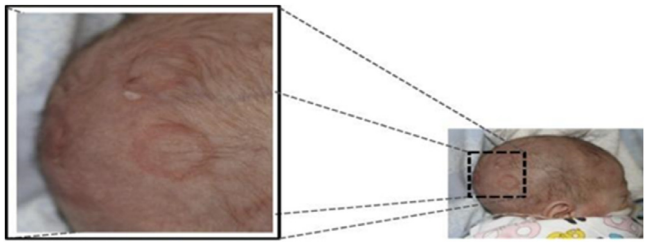


Fig. 9. Marks left behind by the electrode caps [2].

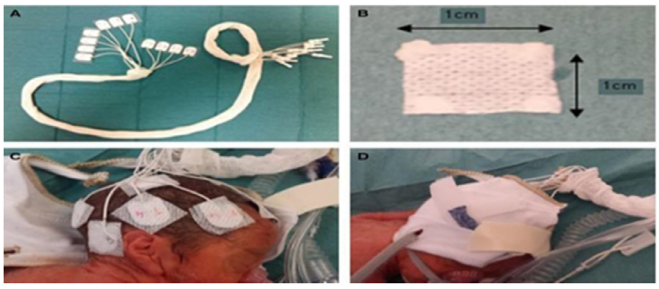


Fig. 10. Textile electrodes for neonates [22].

crucial information can be overlooked. EEG increases the accuracy of the results.

Invasive EEG monitoring methods can have adverse effects on newborns and premature babies.

Wearable devices are becoming more sophisticated with time. This method uses soft conductive textiles as EEG electrodes so that the delicate skin is not harmed in the process. (Hard electrodes are rigid and abrasive in nature). This method is especially useful for neonates; it can also serve the general purpose of unobtrusive long term sensors. The textrodes have many desirable features like flexibility. They can stretch effortlessly and are easy to wash as well. These properties show promise for the future of wearable electronics [22].

The impedance of textile electrodes is comparable with the impedance of Ag/AgCl electrodes over a wide range of frequencies (0.1 Hz–100 kHz) [23]

Compared to hydrogel electrodes, these textrodes or textile electrodes fare better. They are more comfortable to wear, they do not irritate the skin, even when used for a long time. Textrodes can be included into everyday clothing seamlessly. Cleaning the textrodes is convenient

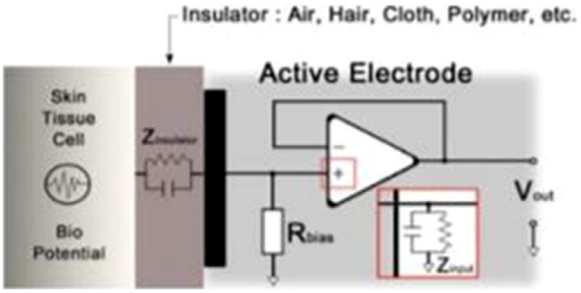


Fig. 11. Capacitive electrode circuit [16].

as well. With thesecoveted properties, it is clear that textile electrodes should be more suitable for long-term uses such as measuring EEG signals for monitoring health [24].

The frequency distribution, spectral coherence and morphology of this method were similar to the other mainstream methods.

Textrodes have shown success in measuring ECG signals [22] Textrodes are polymer foam covered with conductive textile(or a thin layer of silver) [26]. We can make these electrodes by knitting, weaving or embroidering the conductive yarn to the structure or coating, and printing the conductive pastes on the surface of the substrate [26]. Saline water is used as a contact medium in many experiments conducted. This method is primarily focused on the comfort of the patient.

Noise is reduced appreciably when the electrodes are coated with electrolytes. Textile electrodes can capture weak signals with reasonable accuracy; we can perform quantitative analysis on the output. The textrodes dry up after a given amount of time. Hence, the signals are not stable as the medium eventually dries up. This method needs a contact medium (saline water or any other conductive medium) to function. Textile electrodes were introduced to the world as early as 2000, but only a few made it to mainstream electronics as washing of textrodes has always been a problem [27]. Another disadvantage is that these textrodes fail to deliver results for subjects with thick hair. The output is markedly feeble in this case.

Recent research and development in this field have made textrodes more washable [27] and reliable. This has made it possible for patients to use textrodes on a day-to-day basis [24] (Tables 1–3).

2.3. Capacitive electrodes

Richardson proposed the method of gathering biopotentials with capacitors in 1967 [16]. This implied that capacitors could also measure EEG signals. The output was appreciably accurate when compared with the conventional electrodes. As these electrodes are capacitors, they do not require a contact medium. The elimination of contact takes away the disadvantage linked to it(as there is no need for a conductive gel or lengthy procedures to prepare the scalp to pick up the signals.) Matsuo et al(1973) reported this using barium titanate as the insulating material.

The contactless sensor is a fibre-based device whose basis of operation is the electroactive principle [35].

Output analysis made it clear that not only were the alpha waves seen clearly from the non-contact device, but the signals were also virtually identical (with the contact electrodes' output). Signals showed maximum output bandwidth. The strong presence of alpha waves indicated that the non-contact wearable device is sensing valid EEG data.

Unfortunately, low-frequency drift is a common occurrence in non-contact sensors. It can obscure low-frequency EEG features, such as the waves that are produced in a deep sleep state(delta). It can be used for other wavelengths produced.

Capacitive electrodes have various advantages:

1 No leakage in current
2 Gel free procedure
3 Long term monitoring can be inconspicuous [16] The capacitive electrodes have a higher impedance than the impedance offered by the skin. This is to ensure that the electrodes do not pick irrelevant signals. There is a coupling seen between the sensing plate and the skin to achieve a capacitive effect. The coupling capacitance depends mainly on the thickness and the dielectric constant of the material located between the electrode and the subject's skin [16]. Mobile, wearable monitoring systems have become increasingly important in day-to-day life. There are commonmisconceptions about contactless sensors being unreliable and misleading. Research and experimentation have shown that these sensors produce signals comparable to contact and adhesive sensors for EEG applications. Non-contact sensor technology can open the doorway for the EEG technology which was previously confined to hospitals.

2.4. Electrodes integrated with nanotechnology

Invasive electrodes come under Invasive Electroencephalography(iEEG), also known as electrocorticography. This method differs from the other methods, as this technique requires opening the skull of the subject and placing the electrode array on the exposed brain [37]. Due to proximity to the brain, this method records the best quality of signals (taking amplitude and distortion into consideration)

Depth electrodes can be used to monitor the limbic system (deep cortical structures) The amygdala and the hippocampus are important regions in the brain(the amygdala is involved in the processing of emotions and the hippocampus processes declarative or episodic memories).

The medial temporal lobe is the site of epilepsy and seizures(most of the patients have seizures localised in the frontal lobes and the medial temporal lobes)

Despite being a pivotal part of the brain, it is unclear whether their signals can be picked up non-invasively as they are located deep inside the brain. Studying the rhythms of these parts non-invasively is considered to be methodologically challenging [39].

Intracranial electroencephalography is used to measure these regions' brain activity. Intracranial electrodes are used when the scalp recordings do not give us the information needed to localise the part of the brain responsible for the seizures [40]. There are different depths to which the electrodes can go to record signals.

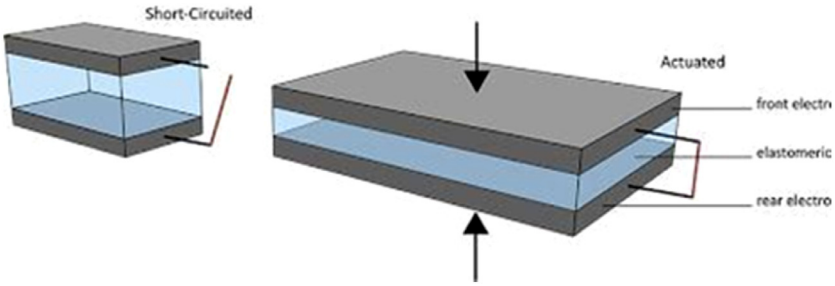


Fig. 12. Electroactive polymers.

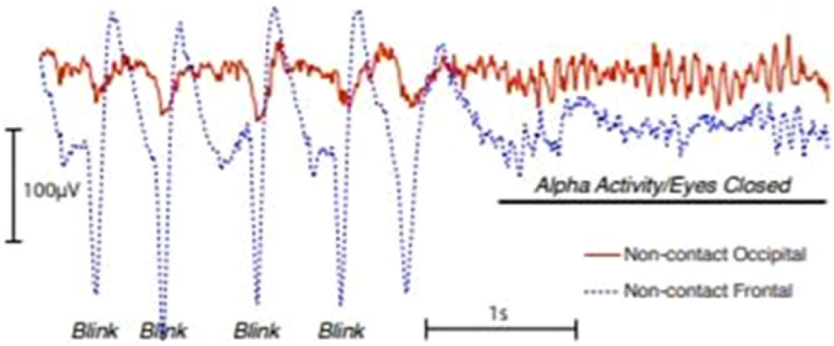


Fig. 13. Experiment showing the signals from the frontal capacitive electrode and occipital capacitive electrode [36].

Table 1
Difference between textile electrode and standard electrode.

Textile electrode	Standard electrode
Textile electrodes are flexible [28]. There are many ways to integrate the conductive material with the fabric. A few of them are: • Knitting:(nylon fibres/spandex fibres are integrated with silver-plated fibres) [28]. • Leaching: PDMS foam with porous titanium [29]. • 3D printing: 3D conductive photopolymer with titanium and gold [30].	Standard electrodes are rigid. There are different kinds of standard electrodes: • Silver/silver chloride gelled electrodes are most commonly used for the biosignal measurement [8]. • Active Dry Single Gold Pin-Based Electrodes [33]. • Silicon material based microneedle electrodes [34].
Textile sensors are best suited for aEEG recordings. Ambulatory EEG is a safe, painless test that records electrical activity produced by the brain [31](usually over a long period)	The signals recorded by standard gel electrodes or invasive electrodes are more accurate than those produced by conductive polymers. Hence, they are used in laboratories and clinics where the accuracy of the signals is more important than the patient's comfort
Textile electrodes show a higher capacitive behaviour when compared to standard electrodes [32]. This is due to the absence of electrolytes. They have a higher skin-electrode impedance when compared to standard electrodes	Standard electrodes have a slightly higher capacitance but they exhibit lower skin-electrode impedance.
One way to reduce skin-electrode impedance is by moisturizing the interface with a hydrogel membrane or saline water [32]	The gel used with standard electrodes decreases the skin-electrode impedance, but irritates the skin. They are abrasive.

Table 2
Difference between invasive and non invasive electrodes.

Invasive electrodes	Non-invasive electrodes
Microelectrodes are 4–5 mm in diameter	Typical diameter of a dry EEG electrode is about 10 mm [52]. A polymer EEG electrode is about 4 mm in diameter.
Subdural electrodes can be 2–5 mm in diameter, they are placed at a distance of 1 cm from one another [45]	A capacitive electrode can be about 7 mm in diameter.
These electrodes are made of non ferromagnetic materials to aid in MRI and MEG scans.	Textile electrodes can be made up of conductive polymers/fabric where the metals are integrated into the polymer/fabric with various methods(knitting, 3D printing etc.)
Subdural electrodes can be made up of stainless steel or platinum-iridium [45].	Metal plates are used as capacitor plates to measure biopotentials arising from the body. Hair acts as the dielectric layer.
Electrodes can be made from rhenium and tantalum as well	
Silicon intracortical probes have the high signal resolutions(due to the proximity with the brain) and low biocompatibility [55]	

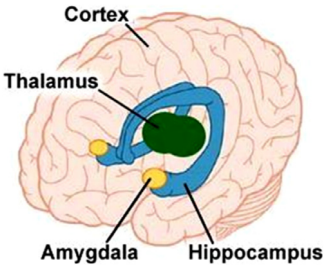


Fig. 14. The amygdala and hippocampus are part of the limbic system [38]. The limbic system is a primitive part of the human brain.

Deep brain recording is the most invasive method, it uses microelectrodes to record signals. The use of microelectrodes to record brain activity was spearheaded by Albe-Fessard and his fellow researchers [42].

These depth electrodes read signals from the region of the brain that they are implanted in. Depth electrodes are capable of detecting the epileptogenic foci(difficult to glean from surface recordings) [44].

Modern advancements in science and technology have helped scientists to generate MRI and CT scans of the affected area of the brain with medically acceptable non-ferromagnetic electrodes.

A semi-invasive method of placing electrodes close to the temporal lobe under fluoroscopy in the vicinity of the foramen ovale is utilised by some practitioners [45]. These electrodes share a resemblance to acupuncture needles.

However, several studies show that this method is as effective as the methods that use scalp electrodes. There are different kinds of subdural electrodes too. They can be placed in a variety of locations, including temporally, sub temporally, frontally and subfrontally [46]. These electrodes have been in existence for the past three to four decades [47,48–50].

Table 3
Schematic analysis of the different kinds of electrodes discussed in this paper.

Dry metal electrodes	Textile electrode	Capacitive electrode	Nanotechnology-based electrodes
Conductive medium is the metal used. Silver is usually used for this purpose	Conductive medium depends on the various ways in which the metal is incorporated into the fabric. For instance, if the metal is knitted, then the conductive medium can be silver plated nylon yarn	The conductors placed on the subject's head act as the plates of the capacitor. The hair and skin in between act as the dielectric medium	Nanomaterials are coated on dry electrodes. These nanomaterials act as the conductive medium
The metal rubs against the skin, and over a period of time, can cause rashes and allergies	Textile electrodes do not irritate the skin as the conductive medium is incorporated into biocompatible fabric	There is no direct contact with the skin. Hence, the chances of skin allergy are at a minimum	Nanomaterials used to coat the electrodes are non-toxic. Nanomaterials are inert, they do not interact with the skin
Dry metal electrodes are not comfortable for everyday use	Textile electrodes are specifically designed for everyday use. They were designed keeping the patients' comfort in mind	Capacitive electrodes can be used for personal health monitoring, completely hidden from view.	Incorporation of nanotechnology has significantly increased the comfort level of the electrodes.
Perspiration, moisture and humidity can change the readings	Textile electrode readings are affected by movement artifact	Perspiration, the thickness of hair and humidity do not affect the readings	Sweat and humidity does not affect signal acquisition. These electrodes perform well even in a mobile environment
The patients can wear the electrodes for about 1.5h [36]	Textile electrodes are long-term electrodes for personal health monitoring	Capacitive electrodes make conformal contact with the scalp. These electrodes are fit for long term use	Nanotechnology can be used as an advantage here as it can be made biocompatible in the molecular level
Abrasive, rigid and not biocompatible Preparation time is maximum when compared to the other non-invasive electrodes	Flexible, light, washable and reusable. Minimal preparation time.	Setting up capacitive electrodes is more time consuming than setting up textile electrodes	Though these electrodes have negligible set up time, manufacturing these electrodes is challenging

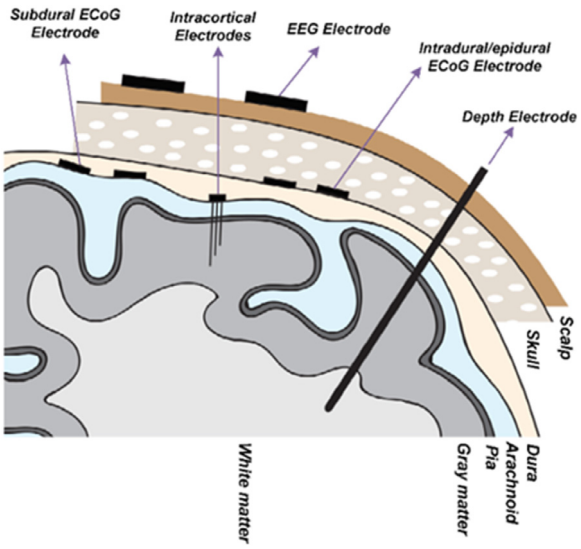


Fig. 15. Different depths at the which the EEG signals can be recorded [41].

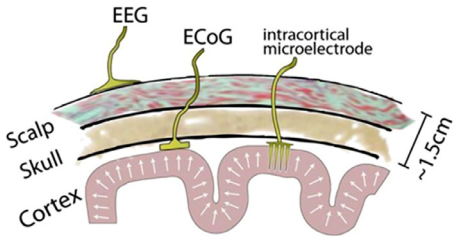


Fig. 16. Cortex penetrating microelectrodes [43].

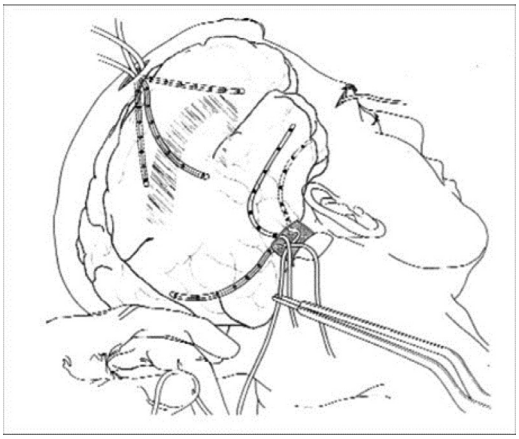


Fig. 17. Tubular subdural electrodes inserted through a posterior temporal burr hole to cover the temporal lobe [50].

Subdural electrodes need larger openings to be placed when compared to depth electrodes. These electrodes are used when the doctors believe that the focal point of the seizure is on the surface

Subdural electrodes, like depth electrodes, are used to improve prognostic and diagnostic accuracy. These electrodes usually are made of metal discs that are placed on silicone rubber or Teflon. These strips can be organised in different configurations: columns, rows, grids etc [51–53].

The electrodes are disc-shaped, usually 4–5 mm in diameter. The material can be a composite of nickel-chromium or platinum-iridium

As shown in Fig. 17, they are planted on the surface of the brain through burr holes. If these electrodes are in the form of grids, they are placed on the surface after a craniotomy is performed [54–56].

MRI imaging can be done with subdural electrodes as well. The composites mentioned above are non-ferromagnetic, thus these electrodes can be used for MRI scans.

Subdural EEG is usually restricted to one hemisphere. There are cases where bi-hemispheric implantations are performed, but it is comparatively rare.

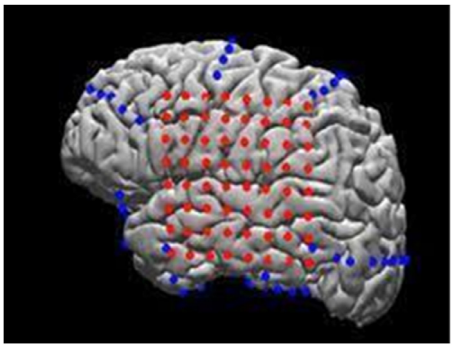


Fig. 18. MRI reconstruction of the patient's brain with the electrodes overlaid [51]

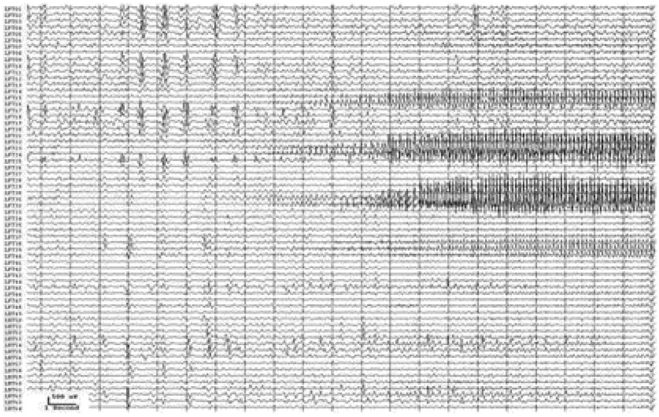


Fig. 19. EG signals from a cortical site(in this case the left rolandic region) [52].



Fig. 20. EEG recording of a person in the wakeful state [55].

Subdural electrodes can cover large areas of the cortex, uninterrupted; thus, easing the process of tracing the seizure. Areas of the brain such as the basal frontal lobe and the basal temporal lobe cannot be implanted with subdural electrodes. These areas carry a greater probability of bleeding due to the presence of draining veins.

Invasive EEG is important as it helps in the expressive mapping of the cortical area of the brain. iEEG recordings have been used for a better part of the century. There have been various technological advancements in this field, but several areas have the scope for development.

Invasive EEG signals are accurate as electrolyte noise is eliminated. This method is advantageous as problems associated with high-impedance can be eliminated. The amplitude of these signals will be higher than surface EEG.

The recordings also show high spatial resolution.

Invasive EEG recordings also have high temporal resolution

Some of the common complications with invasive EEG are:

- increased intracranial pressure
- internal bleeding
- infarction
- post-operative haemorrhage

Surface EEG recordings are susceptible to motion artifacts, the amount of conductive electrolyte present, the thickness of hair(capacitive electrodes can circumvent this problem), perspiration, the environment where the recordings were taken, etc.

The SNR(Signal to Noise Ratio) of surface EEG is less than invasive EEG

Risks associated with surface electrodes are as follows:

- Gels are abrasive and act as irritants
- Dry electrodes cause damage to skin(infants' skin can be damaged while taking the readings)
- Long term use can cause allergies

Surface electrodes pose less risk to the subject when compared to invasive electrodes

Recent developments in nanotechnology have made it possible to have non-invasive electrodes detect epilepsy. CNT(Carbon Nanotube) sensors can be used to measure the electromagnetic waves emanating from the brain. It is a nanotechnology-based sensor that can be used far beyond the hospitals and clinics, in our everyday lives. These are unobtrusive sensors that are comfortable as well.

Non-invasive electrodes have to go through a layer of hair to make successful contact with the skin. Thick hair acts as an insulating layer. Using metal pins to bypass the hair comes with its own risk. Incessant compressions and movement (which causes unintended abrasion) imply that these electrodes need to have good mechanical stability.

Nanotechnology-based electrodes have the potential to overcome all these problems. The electrodes have high conductivity and mechanical stability. The electrodes are chemically inert as well

Multi-Walled CNT can be used to develop EEG sensors as they are good conductors and are more rigid than single- walled carbon nanotubes [15].

G/CNT materials can be used as biosensors. They can be used as EEG electrodes as they are non-toxic and have excellent electrical conductivity. They are stretchable and have superior mechanical properties [53].

Nanowire/polyvinyl butyral(PVB)/melamine sponge (AgPMS) electrodes can be used to measure EEG [54]. It has exceptional mechanical stability and can capture signals even through the insulated hair layer.

Another method to detect seizures is based on the probability of a seizure occurring on a section of the brain. Tunable threshold [55] is used to utilise the detection delay and sensitivity trade-off. This detection system uses scalp EEG that can alert the medical staff about the onset of seizures to greatly increase the speed of clinical diagnosis.

3. Discussions

EEG has always been a crucial component of neurobiology, psychophysiology, BCI(Brain Computer Interfaces) and neurology. Despite playing such a vital role in these fields, the methods of recording these signals is rather rudimentary and limiting. Present-day sensors are limited to hospitals and clinics. This paper aims to change the outlook on the availability of this technology and throw light on alternative methods of sensing these brain signals by using unobtrusive sensors and contactless sensors. It is interesting to note that the alternatives to traditional EEG electrodes produce similar results with very little preparation time. Science has been consistently working towards making this pro-

cess simpler and more accurate. The next step is spreading awareness and making these prototypes the mainstream methods.

4. Conclusions

Non-contact and other unobtrusive monitoring devices will play a crucial part in health systems. Convenient versions of the device help in easing the procedure. They also ensure more accessible diagnostic opportunities. These devices become valuable when supported by a robust and reliable signal analysis infrastructure. Advancements in R&D are making this a reality soon .

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